INTAKE AIR TEMPERATURE (IAT) SENSOR [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0140q1007100

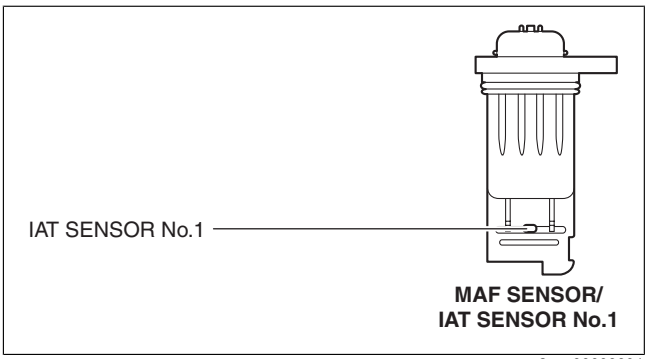
IAT Sensor No.1

Purpose/Function

- Detects the IAT before it passes through the throttle body as basic information (mainly used to determine the fuel injection amount) and sends it to the PCM as an IAT signal.

Construction

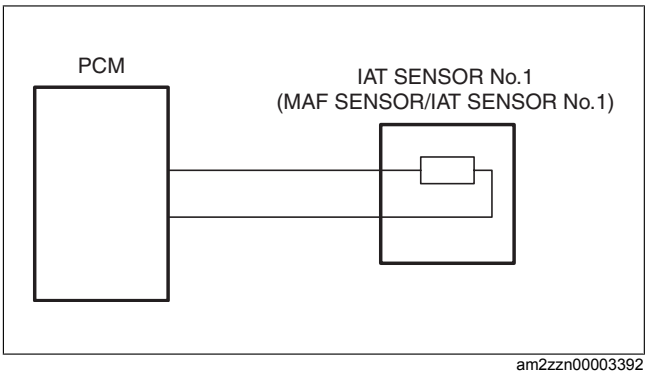
- Built into the MAF sensor.
- A thermistor has been adopted (as the detecting element) for detecting the IAT.



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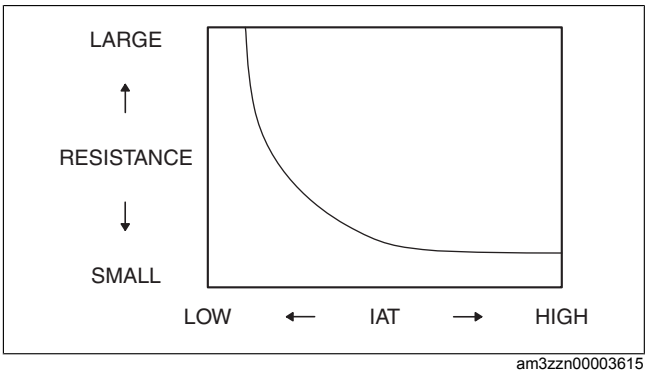
Operation

- Sends the signal, which fluctuates according to the IAT (after passing through the air cleaner), to the PCM.



am2zzn00003392

IAT sensor No.1 characteristic



am3zzn00003615

Fail-safe

DTC No.	Fail-safe function
P0112:00	• Fixes the intake air temperature (for engine control) at 20 °C {68 °F}. • Inhibits the fuel cut control during shift change.
P0113:00	• Fixes the intake air temperature (for engine control) at 20 °C {68 °F}. • Inhibits the fuel cut control during shift change.

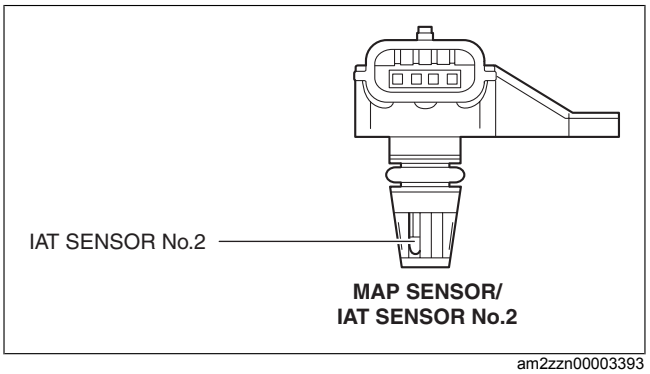
IAT Sensor No.2

Purpose/Function

- Detects the IAT (in intake manifold) after it passes through the throttle body as basic information (mainly used to determine the fuel injection amount) and sends it to the PCM as an intake air temperature signal.

Construction

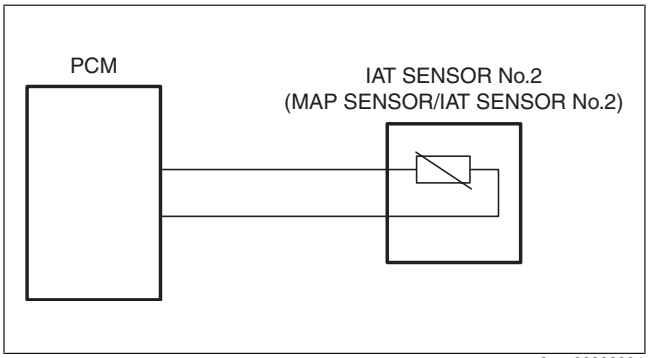
- Built into the MAP sensor.
- A thermistor has been adopted (as the detecting element) for detecting the IAT.



am2zzn00003393

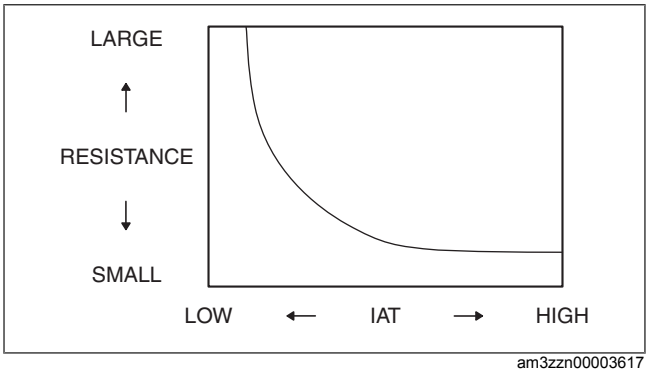
Operation

- Sends the signal, which fluctuates according to the IAT (in the intake manifold), to the PCM.



am2zzn00003394

IAT sensor No.2 characteristic



am3zzn00003617

Fail-safe

- Not applicable